MSc DSAI 2024–2025 — AIEDA323 Biomedical Signal Processing

Computer Labs

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- Please upload on Moodle one separate report per lab assignment.
- Lab reports must include not only results (in the form of figures, tables, etc.) but, more importantly, also comments interpreting and discussing the results.

1 Spectral analysis and filtering

This assignment applies to real signals the spectral analysis and filtering concepts studied in the lectures.

1.1 Abdominal ECG during pregnancy [20%]

The provided data file contains an 8-lead ECG recorded from a pregnant woman at a 500-Hz sampling rate. The first five leads were obtained from the mother's abdominal region, whereas the last three leads were recorded from her thoracic region.

1. Plot the 8 leads. Remark that the 4th abdominal lead is particularly noisy.
2. Using the spectral estimation method of your choice, compute and plot the power spectral density (PSD) of the noisy lead.
3. By observing the estimated PSD, identify the frequency intervals of noise and interference contaminating the signal of interest.
4. Design appropriate frequency filters to reduce noise and interference.
5. Apply the filters to the observed ECG and confirm their effectiveness by analyzing the output signal in both time and frequency domains.

1.2 Atrial fibrillation analysis [80%]

Spectral parameters of the AF ECG have been used as markers of noninvasive atrial activity estimation performance and predictors of therapy outcome. Two such parameters are:

- *Dominant frequency (DF)*, defined as the PSD peak position, f_p, in the interval $[3, 9]$ Hz, the typical AF band. In the literature, low values of this parameter have been linked to high probabilities of spontaneous cardioversion and favorable therapy outcome prediction.

- *Spectral concentration (SC)*, defined as the percentage of signal power localized in a narrow frequency band around f_p. It quantifies the harmonic character or repetitiveness of the atrial activity signal. High values of this parameter have been associated with clean atrial signals, weakly contaminated by residual ventricular components.

This experiment aims at assessing the ability of these spectral parameters to characterize the AF data provided in the module.

1. Create MATLAB functions to compute the DF and SC of the signals given as inputs. As a PSD estimator, use the Welch method with the typical window type and size, overlap factor and number of FFT points found in the literature. Find also in the literature the classical frequency band for SC computation.
2. Compute the DF and SC in lead V1 of all ECG records in the provided AF dataset.
3. With the help of box-and-whiskers plots, compare the distributions of the spectral parameters of the recorded data (`Xva`) versus those of the estimated atrial activity (`Xa`) over the patient population.
4. Design a classifier to predict AF recurrence using DF and SC as features. Show its performance using 5-fold cross-validation.
5. Repeat parts 2 and 4 by averaging, for each patient, the spectral parameters over all precordial leads (V1 to V6).
6. Conclude on the ability of spectral parameters DF and SC to predict AF recurrence in the given dataset.

2 Adaptive noise canceling

This lab assignment deepens the evaluation of multi-reference adaptive noise canceling (MRANC) on synthetic signals and studies its application to real records. The `mranc.m` function coded in the course can be used here.

2.1 Synthetic signals [20%]

1. *Effects of algorithm parameters.* Repeat Exercises 5 and 6 of the course for different values of parameters μ and M:

   (a) $\mu = \{0.005, 0.05\}$, with $M = 0$;

   (b) $M = \{5, 50\}$, with $\mu = 0.01$.

   Draw a table showing the obtained output SIR (dB), SIR improvement (dB) and convergence time (samples) for the 5 tested combinations of μ and M. What can be concluded about the effect of these parameters on the algorithm's performance?

   Note: Remember to plot the relevant signals and to compute SIR values after convergence of the algorithm (steady-state regime), as seen in the course.

2. *Effects of desired signal leakage into the reference signal.* Assume that the reference signals are contaminated by a component of the desired signal attenuated by 10 dB relative to the desired signal power contribution to the primary input. Repeat Exercises 5 and 6 of the course. What can be concluded about the effect of contamination on the algorithm's performance?

2.2 Fetal ECG extraction [20%]

This experiment takes up the pregnant woman's ECG record studied in assignment 1.1. Recall that the first five signals were recorded from abdominal leads, containing a mixture of fetal ECG (FECG) and maternal ECG (FECG), while the last three signals came from thoracic leads, containing MECG only.

1. To obtain a FECG signal free from MECG in each abdominal primary lead, carry out the adaptive cancellation of the MECG interference using the thoracic leads as reference signals. Assume a mixture without time dispersion ($M = 0$).

2. Assess the signal estimation quality:

   (a) Plot and compare the primary inputs and the corresponding output signals. Zoom in the first 2 seconds to help visualize the algorithm's convergence.

   (b) Plot and compare the frequency spectrum (Welch estimate) of the primary inputs and the output signals between 0 and 50 Hz. Consider only the portion of output signals after convergence of the algorithm.

2.3 Atrial activity extraction — single patient [30%]

1. In the AF dataset analyzed in assignment 1.2, consider the precordial leads V1–V6 of the recorded ECG (`Xva`) from the first patient. These signals were recorded from the precordial (chest) region. Plot the 6-lead record. Check that the atrial activity is more clearly present in lead V1.

2. Carry out the adaptive cancellation of the QRST complex taking V1 as primary input and V2–V6 as reference signals. Assume a mixture without time dispersion ($M = 0$).

3. Evaluate the signal estimation quality:

   (a) Plot and compare the primary signal and the filter output signal. Zoom in a 2-second window after convergence to provide a qualitative idea of ventricular activity suppression.

   (b) Plot and compare the frequency spectrum (Welch method) of the primary input and the output signal between 0 et 50 Hz. Consider only the output signal samples after convergence of the adaptive algorithm.

   (c) The DF and SC parameters were introduced in assignment 1.2. Compute these performance indices on the output signal after convergence.

   (d) Compare with the spectral parameters of the atrial activity signals provided in matrix `Xa`, obtained by the spatio-temporal cancellation (STC) method in experiment 1.2.

2.4 Atrial activity extraction — full database [30%]

1. Repeat experiment 2.3 for the full patient dataset. When evaluating performance (part 3), just compute spectral parameters DF and SC without plotting results.

2. In terms of spectral parameters, compare MRANC with the STC method (matrix `Xa`). Conclude.

3. [BONUS] Following the guidelines of experiment 1.2, test the ability of the spectral parameters of the MRANC atrial estimate to predict AF recurrence in the given dataset.

3 Blind source separation

The performance of the BSS approach both in synthetic and real signals is evaluated in this lab assignment. To perform PCA, use the `bss_pca.m` function coded in the course. To perform ICA, first download and install the RobustICA algorithm.

3.1 Synthetic signals [20%]

1. Repeat Exercises 5–7 of the course after reducing the desired signal power contribution to the first sensor to 0.01. Conclude.
2. *Effects of desired signal leakage into the reference signals.* Retrieve the observed signals from experiment 2.1-2, where the reference signals were contaminated by components of the signal of interest. Repeat Exercises 5–7 of the course with these observations. Conclude.

3.2 Fetal ECG extraction [20%]

Retrieve the ECG recording of a pregnant woman.

1. Perform the separation of the maternal and fetal heartbeat sources with PCA and ICA, taking the 8 recorded signals as observations.
2. Assess the separation quality obtained with PCA and ICA:
   (a) Plot (separately for each method) the obtained sources.
   (b) Visually identify the fetal ECG (FECG) sources. Note that there may be more than one source of fetal cardiac activity.
   (c) Compute the fetal source contribution to all recordings.
   (d) Plot and compare the abdominal signals and the estimated FECG contributions.
   (e) Plot and compare the frequency spectrum (Welch estimate) of the abdominal signals and the estimated FECG contributions between 0 and 50 Hz.
3. Compare these results with those obtained by the MRANC technique in assignment 2.2.

3.3 Atrial activity extraction — single patient [30%]

1. Retrieve the first patient's record of the AF ECG dataset analyzed in assignments 1.2 and 2.3. Consider the precordial leads V1–V6 of the recorded ECG (`Xva`).
2. Perform the source separation via PCA and ICA from the 6 thoracic leads.
3. Assess the separation quality:
   (a) Plot (separately for each method) the obtained sources.
   (b) Visually identify the atrial activity (AA) sources. Remark that the number of AA sources estimated by PCA and ICA may be different.
   (c) Plot de PSD of the estimated sources.
   (d) Automatically identify the atrial source as the signal with the highest SC among the estimated components with DF in the AF interval $[3, 9]$ Hz. Does automatic identification coincide with visual inspection of the atrial source?
   Note: The DF and SC parameters were introduced in assignment 1.2.

(e) Compute the contribution of the AA sources to lead V1.

(f) Plot the signal observed on lead V1 and superimpose the estimated AA contribution.

(g) Plot and compare the frequency spectrum (Welch estimate) of lead V1 and the estimated AA contribution between 0 and 25 Hz.

(h) Compute the DF and SC performance indices on the atrial signal estimated in lead V1.

4. Compare these results (plots and spectral parameters) with those obtained by MRANC and STC in experiment 2.3.

3.4 Atrial activity extraction — full database `[30%]`

1. Repeat experiment 3.3 for the full patient dataset, taking into account the following changes:

   (a) Assume that all atrial activity is due to a single atrial source that can be identified automatically based on the procedure described in the experiment 3.3.

   (b) When evaluating performance (part 3), just retain spectral parameters DF and SC of the selected atrial source without plotting results.

2. Compare PCA, ICA, MRANC and STC (`Xa`) in terms of spectral parameters. Conclude.

3. `[BONUS]` Following the guidelines of experiment 1.2, verify the ability of the spectral parameters of the atrial sources estimated by BSS to predict AF recurrence in the given dataset.